Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/12/2007
PI Name:	Williams, Gary A Ph.D.		
Project Title:	Simultaneous superfluid density and Casimir thinning measurements in thin films of He-4 near the lambda point		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gaw@ucla.edu	**Fax:**	FY 310-206-5668
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-8536
Organization Name:	University of California - Los Angeles		
PI Address 1:	Physics and Astronomy		
PI Address 2:			
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1547	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Penanen, Kontstantin (Jet Propulsion Laboratory) Chui, Talso (Jet Propulsion Laboratory)		
Grant/Contract No.:	GSN 4534		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

Task Description:	The effective forces associated with fluctuations in restricted geometries, broadly called Casimir forces, have recently been directly observed both in vacuum and in several condensed matter systems. This effect is particularly strong near critical points, where the correlation length diverges and exceeds the characteristic size of the system, such as the thickness of a film. In the superfluid transition in 4He, the Casimir forces cause the film to thin in a very narrow temperature range about the critical point. We propose a series of experiments designed to measure simultaneously the film thickness and superfluid density in saturated films of 4He. To avoid complications due to capillary condensation and to possible pinning of vortices by surface defects, we propose a novel design of the capacitor that measures the film thickness, with the expected RMS roughness of the silicon surfaces covered with helium being on the order of 0.3 nm. To mitigate thickness changes due to the fountain effect a temperature-stabilized system employing thermometry and temperature control with sub-nanokelvin resolution will be used. We will use third sound, a low-frequency hydrodynamic mode propagating along the substrate surface, as a simultaneous probe of the superfluid density. Measure simultaneously the film thickness and superfluid density of thick films (40-100 nm) of liquid 4He on a silicon substrate, over a narrow temperature range (10 mK) above and below the superfluid onset temperature. The Casimir force that causes helium films to thin at the transition temperature is poorly understood, with an existing discrepancy between theory and experiment of about a factor of fifty. Our measurements on a nearly perfect substrate will be a definitive test of the theory, and the superfluid density measurement will resolve the location of the superfluid onset temperature relative to the film thinning, also a major test of the theory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will contribute to a basic understanding of the properties of superfluid phase transitions, testing the accuracy of renormalization-group methods of statistical mechanics.
Task Progress:	RESEARCH PROGRESS FOR FINAL REPORTING The experimental work has continued, and we have now successfully observed a third sound signal near the superfluid transition temperature. The speed of the third sound to as close as 60 mK from the transition is in good agreement with theoretical calculations. We are currently continuing the experimental run to push even closer to the transition and find the critical Kosterlitz-Thouless onset point for 2D superfluidity. We will also monitor the helium film thickness to observe how the Casimir thinning effect correlates with the third sound observations.
Bibliography Type:	Description: (Last Updated: 10/15/2007)
Articles in Peer-reviewed Journals	Williams GA. "Anisotropic Kosterlitz-Thouless transition induced by hard-wall boundaries." Physical Review B. 2006 Jun;73(21):214531. http://dx.doi.org/10.1103/PhysRevB.73.214531 , Jun-2006